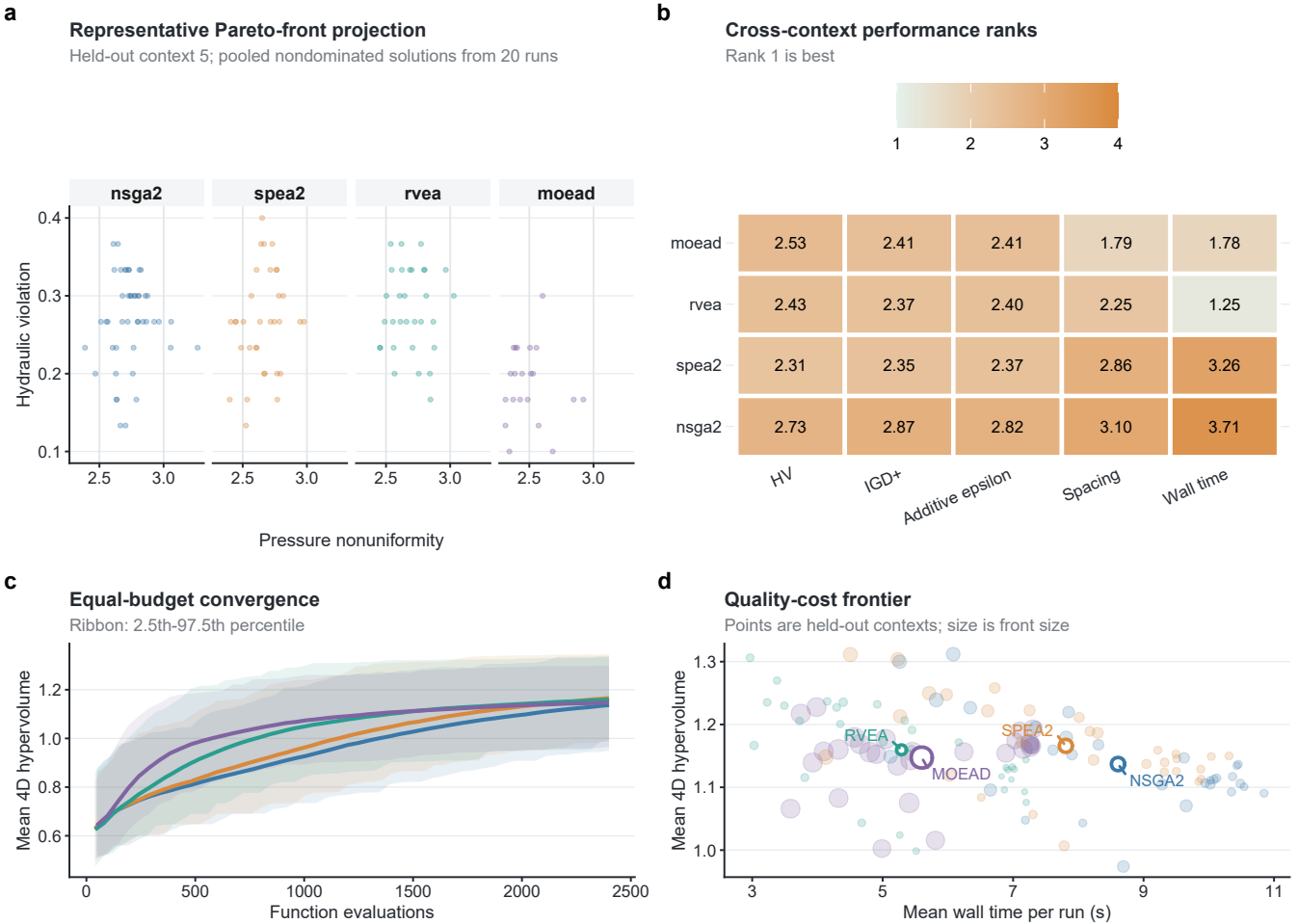

a
Representative Pareto-front projection
Held-out context 5; pooled nondominated solutions from 20 runs
nsga2
spea2
rvea
moead
Hydraulic violation
0.4
0.3
0.2
0.1
2.5
3.0
Pressure nonuniformity
b
Cross-context performance ranks
Rank 1 is best
1
2
3
4
moead 2.53 2.41 2.41 1.79 1.78
rvea 2.43 2.37 2.40 2.25 1.25
spea2 2.31 2.35 2.37 2.86 3.26
nsga2 2.73 2.87 2.82 3.10 3.71
HV
IGD+
Additive epsilon
Spacing
Wall time
c
Equal-budget convergence
Ribbon: 2.5th-97.5th percentile
Mean 4D hypervolume
1.2
1.0
0.8
0.6
0
500
1000
1500
2000
2500
Function evaluations
d
Quality-cost frontier
Points are held-out contexts; size is front size
Mean 4D hypervolume
1.3
1.2
1.1
1.0
RVEA
MOEAD
SPEA2
NSGA2
3
5
7
9
11
Mean wall time per run (s)